

ALPACA HACKATHON 2026

Catalvst Router

An autonomous paper-trading agent
where AI finds a signal and deterministic
systems protect the capital.

Make the model useful, not **all-powerful.**

AI proposes direction and confidence.
Deterministic code owns every dollar of
risk.

OBSERVE

Dynamic universe

Liquid equities, IEX prices, and fresh market context.

DECIDE

Long / Short / Abstain

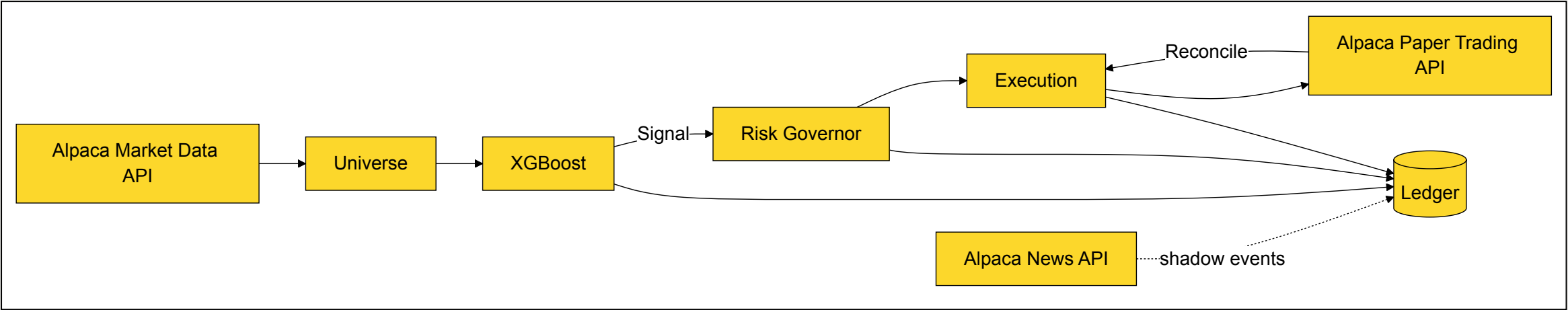
XGBoost scores completed 15-minute bars.

GOVERN

Risk first

The model cannot size, submit, or bypass a veto.

One narrow path to an order.



The XGBoost model can propose a trade. Only the Risk Governor can approve an Alpaca order.

BOOSTED TREES	MAX DEPTH	POINT-IN-TIME INPUTS	PREDICTION HORIZON
350	4	16	4H

The classifier estimates a positive four-hour move. A probability of 52% or higher indicates enough upside edge for a long position. A probability of 48% or lower indicates the corresponding downside edge for a short position. The 48-52% band abstains from weak signals. This narrow symmetric gate permits more paper trading opportunities while deterministic risk controls retain final authority.

Alpaca is the market interface and the **source of truth.**

MARKET DATA API

IEX bars, quotes, screeners

Point-in-time model features plus freshness and spread gates.

NEWS API

Catalyst evidence

Symbol news feeds a shadow-only Bedrock event extractor.

PAPER TRADING API

Brackets and reconciliation

Broker-hosted exits; account, order, and P&L authority.

ALPACA-PY + CLI

Typed runtime and safe demo

Python worker executes; CLI supports read-only diagnostics and dry runs.

Every trade must earn permission to exist.

MARKET GATES

Freshness, spread, hours

Bad or stale data resolves to no trade.

PORTFOLIO LIMITS

1% risk per trade

Six positions max, 5% total stop-risk.

FAILURE CONTAINMENT

Brackets + circuit breakers

Risk halt at 4% daily loss; kill at 12% drawdown.

Auditable by design: immutable decisions, idempotent order claims, and Alpaca reconciliation after every restart.

Paper-trading performance.

NET P&L

+\$115.01

RETURN

+0.12%

FILLED ORDERS

42

MAX DRAWDOWN

-0.19%

Measured against Alpaca's \$100,000 baseline: the account ended September 3 at \$100,115.01.

Peak before options. The account reached \$100,309.97 on September 2, or +\$309.97 from baseline. The first option session on September 3 lost approximately \$194 after stale indicative quotes triggered a risk halt and flatten. Option execution is now disabled.

CATALYST ROUTER

Intelligence with boundaries.

We are not asking an LLM to trade. We are building a verifiable system that uses models where they help and controls where they matter.

AI proposes. Deterministic code disposes. Alpaca reconciles.